ALIZA BEVERAGE

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RESEARCH INTERESTS

The formation and evolution of galaxies, stars and stellar populations, star-formation quenching, origins of the elements, chemical evolution

PROFESSIONAL APPOINTMENTS

Hubble Fellow, Carnegie Observatories — September 2025 – Present
Carnegie-Princeton Fellow, Princeton University — September 2025 – Present

EDUCATION

University of California Berkeley, *Ph.D., Astrophysics* — August 2025
Thesis: Charting the Chemical Compositions of Massive Galaxies Across Cosmic Time
Advisors: Mariska Kriek and Dan Weisz

University of California Berkeley, *Master of Arts, Astrophysics* — August 2020

University of Minnesota Twin Cities, *Bachelor of Science, Physics and Astrophysics* — June 2018
Advisor: Claudia Scarlata

AWARDS

NHFP Hubble Fellowship	2025–2028
Carnegie-Princeton Fellowship	2028–2030
NSF Graduate Research Fellowship	2019–2024
Robert J. Trumpler Graduate Student Excellence Award	2024
In recognition of academic excellence and outstanding record of involvement in the Berkeley department or wider astronomical community.	
H2H8 Fellowship ($10,000)	2022
AAS International Travel Grant ($6,000)	2021, 2023, 2024
Outstanding Astronomy Graduate Student Instructor	2021
Maria Cranor Fellowship ($18,000)	2019–2025
Astronaut Scholarship ($10,000)	2018–2019

GRANTS AND FUNDING

***Total funding awarded:* $500,000**

JWST GO 11451 (PI), STScI	$310,000, Cycle 5
JWST GO 11532 (co-PI with Chloe Cheng), STScI	$72,000, Cycle 5
JWST GO 5629 (co-PI with Mariska Kriek and Chloe Cheng), STScI	Cycle 3
JWST GO 2110 (co-PI with Mariska Kriek), STScI	Cycle 1
NASA Hubble Fellowship, research budget	$79,500, 2025–2028
Carnegie-Princeton Fellowship, research budget	$38,500, 2028–2030

OBSERVING PROGRAMS AS PI AND CO-PI

112 *JWST* hrs as PI/co-PI

JWST, NIRSpec (GO 11451) — 43.9 hrs, Cycle 5
How Fast Is Too Furious? Ultradeep Spectroscopy of a Fossil from Cosmic Dawn

JWST, NIRCam (co-PI with Chloe Cheng; GO 11532) — 5.4 hrs, Cycle 5
It's the Little Things that Count: Unveiling Faint Merger Signatures and Tiny Companions around Massive Quiescent Galaxies at $z \sim 2$

JWST, NIRSpec (co-PI with Mariska Kriek and Chloe Cheng; GO 5629) — 40.3 hrs, Cycle 3
Extremely Deep Spectroscopy of Quiescent Galaxies at $z \sim 0.7$: A Direct Measurement of the Stellar Initial Mass Function Beyond the Low-Redshift Universe

JWST, NIRSpec (co-PI with Mariska Kriek; GO 2110) — 22.7 hrs, Cycle 1
Ultra-Deep Continuum Spectroscopy of Quiescent Galaxies at $1.0 < z < 2.5$: Chemical Abundances and Stellar Kinematics

Magellan, LLAMAS — 9 nights, 2025B–2027A
The Anemic Survey: Spatially Resolving the Chemical Abundances, Kinematics, and IMF of Quiescent Galaxies

OBSERVING PROGRAMS AS CO-INVESTIGATOR

JWST, NIRSpec and NIRCam (PI: Katherine Whitaker; GO 11883) — 32.9 hrs, Cycle 5
Opening a New Window on Globular Cluster Formation with Strong Lensing at $z = 2$

JWST, NIRSpec (PI: Elham Eftekhari; GO 10660) — 38.2 hrs, Cycle 5
Unlocking the Power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function in Quiescent Galaxies at Intermediate-z

JWST, NIRSpec and NIRCam (PI: Nathan Sandford; GO 10246) — 131.3 hrs, Cycle 5
Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

JWST, NIRSpec (PI: Mariska Kriek; GO 8317) — 7.9 hrs, Cycle 4
The Return of the Giants: Constraining the TP-AGB Phase across Cosmic Time

Keck, LRIS and MOSFIRE (PI: Mariska Kriek) — 10 nights, 2020A–2021B
The Heavy Metal Survey: The chemical enrichment, star-formation and assembly histories of $z \sim 1.4 - 2.3$ quiescent galaxies

TEACHING & ADVISING

Research mentor to PhD student Rana Osman, Princeton University — 2026−Present
Publication: *The chemical compositions of massive galaxies at $z \sim 2$ (in prep.)*

Research mentor to Yilun Ma, UC Berkeley (now PhD student at Princeton) — 2020–2022
Publication: *Two Transitional Galaxies with AGN-Driven Outflows at $z \sim 2$ (in prep.)*

*__Graduate Student Instructor__, UC Berkeley 7B: Introduction to Astrophysics II — Spring 2020

Graduate Student Instructor, UC Berkeley 7A: Introduction to Astrophysics I — Fall 2019

**Awarded Outstanding Astronomy GSI*

OUTREACH, SERVICE & DEI

Referee The Astrophysical Journal, Astronomy & Astrophysics — 2021–Present

POWER Coordinator — 2022–2025
POWER is a graduate student-led mentoring program and workshop series for Bay Area community college students interested in the physical sciences. I organized and led over 5 workshops that reached over 100 students and included financial aid and support network presentations, career and transfer student panels, and lab tours.

MPS Scholars Mentor 2023
Professional development mentorship to a Berkeley undergrad

MPS Scholars Retreat Spring 2024
Graduate student panelist at professional development retreat for Berkeley undergrads

POWER Mentor 2021
Professional development mentorship to a Bay Area community college student

Climate & DEI Committee Rep 2020–2021
Graduate student representative on the Berkeley Astronomy DEI committee

Respect is Part of Research, *Facilitator* 2021

Grad Student Postdoc Seminar, *Organizer* 2022–2025

COLLOQUIA & SEMINARS

Penn State	Colloquium, 2027
University of Minnesota	Colloquium, 2026
Texas A&M	Colloquium, 2026
UCLA	Colloquium, 2026
Leiden University	Colloquium, 2026
UC Davis	Cosmology Seminar, 2026
Carnegie Observatories	Lunch Seminar, 2024
Ohio State University	Galaxy Seminar, 2024
Washington State University	Colloquium, 2023
Princeton University	Galread, 2022

CONFERENCE TALKS

Invited:

Liguria, Italy, The Present and Future of High-Redshift Science with JWST	2027
Bern, Switzerland, ISSI: Clumpy Star Formation	2027
UIUC, USA, The End of Star Formation	2026
Sesto, Italy, Cosmic Nuggets II	2026
Sesto, Italy, The Growth of Galaxies in the Early Universe	2026
Northwestern, USA, CIERA Fellows at 15 Conference	2025
Leiden, Netherlands, Lorentz Workshop on Big Galaxies, Big Problems	2025
Yale, USA, The Inaugural Tinsley Workshop	2024
UC Santa Cruz, USA, Galaxy Workshop	2023

Contributed:

Grindelwald, Switzerland, Forging Light: Interpreting Observations of High-Redshift Galaxies	2027
Tokyo, Japan, The Universe in Full Bloom	2026
Sirolo, Italy, AGN Feedback and Star Formation Across Cosmic Scales and Time	2024
Cape Town, South Africa, IAU: The First Chapters of Our Cosmic History with *JWST*	2024
Tenerife, Spain, A Life Devoted to Stellar Populations	2023
Berkeley, USA, Keck Science Meeting	2023
Catania, Italy, Charting the Metallicity Evolution History of the Universe	2022
Baltimore, USA, STScI Multi-Object Spectroscopy Workshop	2021

OPEN SOURCE SOFTWARE

`alf`α: Chemical abundance fitting code for old stellar systems (A. Beverage 2024)
A Python adaptation of `alf` *(Conroy et al. 2018)*

PUBLICATION LIST

6 first-author, 7 second- and third-author papers with significant contribution (ADS library)

First-author:

6. **Beverage, A. G.** & J. Johnson, *Chemical Consequences of a Top-Heavy IMF*, in prep.
5. **Beverage, A. G.**, D. Weinberg, M. Kriek et al., *Are Nucleosynthetic Yields Universal? Interpreting the Multi-Elemental Abundances of Quiescent Galaxies over Cosmic Time Using Milky Way Stars*, Accepted to ApJ, (arXiv)
4. **Beverage, A. G.**, M. Slob, M. Kriek et al., *Carbon and Iron Deficiencies in Quiescent Galaxies at $z = 1$–3 from JWST-SUSPENSE: Implications for the Formation Histories of Massive Galaxies*, 2025, ApJ, 979, 249, DOI: 10.3847/1538-4357/ad96b6
3. **Beverage, A. G.**, M. Kriek, K. A. Suess et al., *The Heavy Metal Survey: The Evolution of Stellar Metallicities, Abundance Ratios, and Ages of Massive Quiescent Galaxies since $z \sim 2$*, 2024, ApJ, 966, 234, DOI: 10.3847/1538-4357/ad372d
2. **Beverage, A. G.**, M. Kriek, C. Conroy et al., *From Carbon to Cobalt: Chemical Compositions and Ages of $z \sim 0.7$ Quiescent Galaxies*, 2023, ApJ, 948, 140, DOI: 10.3847/1538-4357/acc176
1. **Beverage, A. G.**, M. Kriek, C. Conroy et al., *Elemental Abundances and Ages of $z \sim 0.7$ Quiescent Galaxies on the Mass-Size Plane: Implication for Chemical Enrichment and Star Formation Quenching*, 2021, ApJL, 917, L1, DOI: 10.3847/2041-8213/ac12cd

Second- and Third-author:

7. Hamadouche, M. L., J. Antwi-Danso, **A. G. Beverage**, and K. E. Whitaker, *A Spectral Atlas for Quiescent Galaxies*, 2026, Res. Notes AAS, 10, 133, DOI: 10.3847/2515-5172/ae7077
6. Suess, K. A., **A. G. Beverage**, M. Kriek et al., *Cold Gas in a Post-Starburst Pair in the $z \sim 1.4$ HeavyMetal Survey: Major Mergers as a Pathway to Retain Gas in Quenched Galaxies*, 2025, ApJ, 993, 158, DOI: 10.3847/1538-4357/ae0933
5. Gountanis, N. M., D. H. Weinberg, **A. G. Beverage** et al., *Modeling the Ages and Chemical Abundances of Elliptical Galaxies*, 2025, ApJ, 985, 184, DOI: 10.3847/1538-4357/adc4e6
4. Cheng, C. M., M. Kriek, **A. G. Beverage** et al., *Ages and Metallicities of Quiescent Galaxies: Confronting Broad-Band (UVJ) Colours with Stellar Absorption Lines*, 2025, MNRAS, 540, 1527, DOI: 10.1093/mnras/staf806
3. Cheng, C. M., M. Kriek, **A. G. Beverage** et al., *Age and Metal Gradients in Massive Quiescent Galaxies at $0.6 < z < 1.0$: Implications for Quenching and Assembly Histories*, 2024, MNRAS, 532, 3604, DOI: 10.1093/mnras/stae1739
2. Slob, M., M. Kriek, **A. G. Beverage** et al., *The JWST-SUSPENSE Ultradeep Spectroscopic Program: Survey Overview and Star-Formation Histories of Quiescent Galaxies at $1 < z < 3$*, 2024, ApJ, 973, 131, DOI: 10.3847/1538-4357/ad65ff
1. Kriek, M., **A. G. Beverage**, S. H. Price et al., *The Heavy Metal Survey: Star Formation Constraints and Dynamical Masses of 21 Massive Quiescent Galaxies at $z = 1.3$–2.3*, 2024, ApJ, 966, 36, DOI: 10.3847/1538-4357/ad2df9